

Practical 13 - GLMs for grouped binary and count outcomes

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `medpar.csv`

In this practical we will explore models for a count variable and look at the two approaches described in the lecture for handling overdispersion. We will use publicly available data on length of hospital stay from Arizona in the US.

In our analyses we will focus on the following variables.

Variable	Description
<code>los</code>	Length of hospital stay, in days
<code>age</code>	Age group (factor variable)
<code>type1</code>	Binary variable indicating elective admission
<code>type2</code>	Binary variable indicating urgent admission
<code>type3</code>	Binary variable indicating emergency admission

Note that admission type is here described using three binary indicator variables, rather than as a 3-level categorical variable.

Aims

1. Understand how to fit models to count data using the `glm` command.
2. Understand how to compare such models.
3. Understand the concept of overdispersion, and how to identify it.

Question 1

Explore the length of stay variable. How is it distributed? What are the minimum and maximum values in this dataset?

Construct a 95% confidence interval for the mean length of stay using the standard approach taught in the Analytical Techniques course.

SOLUTIONS

```
medpar <- read.csv("medpar.csv")
summary(medpar)
```

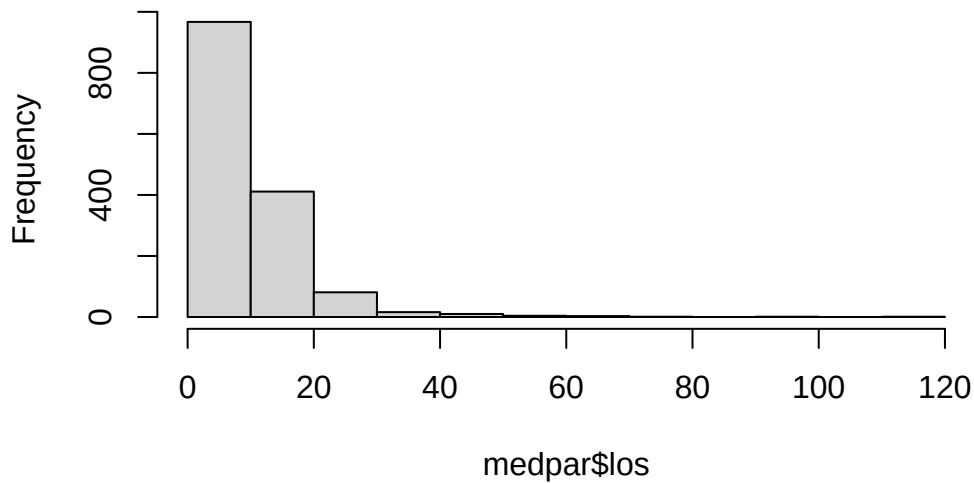
provnum	died	white	hmo
Min. :30001	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:30013	1st Qu.:0.0000	1st Qu.:1.0000	1st Qu.:0.0000
Median :30036	Median :0.0000	Median :1.0000	Median :0.0000
Mean :30107	Mean :0.3431	Mean :0.9151	Mean :0.1599
3rd Qu.:30080	3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:0.0000
Max. :32003	Max. :1.0000	Max. :1.0000	Max. :1.0000
los	age80	age	type1
Min. : 1.000	Min. :0.0000	Min. :1.000	Min. :0.0000
1st Qu.: 4.000	1st Qu.:0.0000	1st Qu.:4.000	1st Qu.:1.0000
Median : 8.000	Median :0.0000	Median :5.000	Median :1.0000
Mean : 9.854	Mean :0.2207	Mean :5.235	Mean :0.7585
3rd Qu.:13.000	3rd Qu.:0.0000	3rd Qu.:6.000	3rd Qu.:1.0000
Max. :116.000	Max. :1.0000	Max. :9.000	Max. :1.0000
type2	type3		
Min. :0.0000	Min. :0.00000		
1st Qu.:0.0000	1st Qu.:0.00000		
Median :0.0000	Median :0.00000		
Mean :0.1773	Mean :0.06421		
3rd Qu.:0.0000	3rd Qu.:0.00000		
Max. :1.0000	Max. :1.00000		

```
var(medpar$los)
```

```
[1] 78.02022
```

```
hist(medpar$los)
```

Histogram of medpar\$los



The los variable has a highly positively skewed (non-normal) distribution. It ranges from 1 to 116 days, with an observed mean of 9.85 and median of 8. The variance (78.0) is much larger than the mean.

The standard approach taught in Analytical Techniques is based on assuming the variable is normally distributed. Since the sample size here is very large, we can use the sample mean ± 1.96 standard errors:

```
n <- dim(medpar)[1]
mean(medpar$los)-1.96*(sd(medpar$los) / sqrt(n))
```

```
[1] 9.406427
```

```
mean(medpar$los)+1.96*(sd(medpar$los) / sqrt(n))
```

```
[1] 10.30193
```

Or alternatively, we could use `t.test`:

```
t.test(medpar$los)
```

One Sample t-test

```

data:  medpar$los
t = 43.136, df = 1494, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
  9.406072 10.302289
sample estimates:
mean of x
  9.854181

```

The 95% CI extends from 9.406 to 10.302. This CI is valid provided the variable is normally distributed. Even if the variable is not normally distributed the coverage of the CI will be close to 95% if the sample size is sufficiently large. This is a consequence of the central limit theorem. Here the coverage should be reasonable since the sample size is quite large.

Question 2

Use the `glm` function to fit a Poisson model for length of stay, with no covariates.

What does the constant coefficient in this model represent? Construct a 95% confidence interval for the mean length of stay.

What can you infer from the reported deviance statistic?

SOLUTIONS

```

losPoisson <- glm(los~1, data=medpar,family=poisson)
summary(losPoisson)

```

Call:

```
glm(formula = los ~ 1, family = poisson, data = medpar)
```

Coefficients:

```

              Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.287896    0.008239   277.7   <2e-16 ***
---

```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(Dispersion parameter for poisson family taken to be 1)

```

Null deviance: 8901.1  on 1494  degrees of freedom
Residual deviance: 8901.1  on 1494  degrees of freedom

```

AIC: 14618

Number of Fisher Scoring iterations: 5

The estimated constant coefficient in this model (2.288) represents the log of the mean length of stay. We can see this by calculating $e^{2.288} = 9.854$. We can calculate a 95% CI based on the estimate and the SE with:

```
exp(2.287896-1.96*0.008239)
```

```
[1] 9.696331
```

```
exp(2.287896+1.96*0.008239)
```

```
[1] 10.0146
```

The residual deviance (8901.1) is substantially larger than the residual degrees of freedom (1494). This strongly suggests overdispersion.

Question 3

Load (and install if necessary) the `sandwich` package. Then follow the example from the lecture to calculate the sandwich (robust) standard error for the coefficient in the Poisson model you fitted before. Use this to construct a 95% confidence interval.

Compare and contrast the 95% confidence intervals in questions 1, 2 and 3. Which is most appropriate? Which is least appropriate?

SOLUTIONS

```
library(sandwich)
vcovHC(losPoisson)
```

```
              (Intercept)
(Intercept) 0.0005377935
```

```
#robust/sandwich SE is sqrt
sqrt(vcovHC(losPoisson))
```

```
(Intercept)
(Intercept) 0.02319038
```

```
#95% CI is then
coef(losPoisson)[1]-1.96*sqrt(vcovHC(losPoisson))
```

```
(Intercept)
(Intercept) 2.242443
```

```
coef(losPoisson)[1]+1.96*sqrt(vcovHC(losPoisson))
```

```
(Intercept)
(Intercept) 2.333349
```

```
#and then exp() to get 95% CI for the mean of los
exp(coef(losPoisson)[1]-1.96*sqrt(vcovHC(losPoisson)))
```

```
(Intercept)
(Intercept) 9.416304
```

```
exp(coef(losPoisson)[1]+1.96*sqrt(vcovHC(losPoisson)))
```

```
(Intercept)
(Intercept) 10.31242
```

The estimated mean is the same using all three approaches. The 95% CI is much narrower when making the Poisson assumption (9.696 to 10.015) than using either of the other two approaches (9.406 to 10.302 and 9.416 to 10.312), which are very similar to each other.

The difference between the 95% CIs in questions 2 and 3 can be attributed to overdispersion in `los`: the deviance statistic is much larger than its d.f. This makes sense given the context; the underlying severity of incoming patients' conditions will vary, leading to more variability than predicted by the Poisson distribution. So, the approach in question 2 is not appropriate.

The approach in question 1 takes no account of the skewness of the distribution of `los` (the 95% CI here is symmetric about the mean on an arithmetic scale), so the approach in question 3 is probably to be preferred (here the 95% CI is symmetric about the mean on a multiplicative scale), but (because of the large sample size and the central limit theorem) the approach in question 1 is also acceptable.

Question 4

We will now explore which (if any) factors are related to length of hospital stay. First fit a Poisson model to assess whether length of hospital stay is related to the type of admission.

Interpret each of the parameter estimates in your model.

SOLUTIONS

Note that we only need to include two of the indicator variables (we cannot include all three). We'll include `type2` and `type3`, so that type 1 becomes the reference level:

```
losPoissonType <- glm(los~type2+type3, data=medpar,family=poisson)
summary(losPoissonType)
```

Call:

```
glm(formula = los ~ type2 + type3, family = poisson, data = medpar)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.178233	0.009993	217.98	<2e-16 ***
type2	0.237344	0.020902	11.36	<2e-16 ***
type3	0.725361	0.025903	28.00	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 8901.1 on 1494 degrees of freedom
Residual deviance: 8183.6 on 1492 degrees of freedom
AIC: 13905

Number of Fisher Scoring iterations: 5

```
#to test the hypothesis that the (log)
#means are the same in the three groups, we can compare to our
#earlier model:
anova(losPoisson,losPoissonType)
```

Analysis of Deviance Table

```

Model 1: los ~ 1
Model 2: los ~ type2 + type3
      Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         1494      8901.1
2         1492      8183.6  2    717.51 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

For interpretation it's nicer to exponentiate the coefficients and the CI limits. We can that programmatically with:

```
exp(coef(losPoissonType))
```

```

(Intercept)      type2      type3
  8.830688    1.267877    2.065477

```

```
exp(confint.default(losPoissonType))
```

```

              2.5 %   97.5 %
(Intercept) 8.659413 9.005350
type2       1.216985 1.320897
type3       1.963233 2.173046

```

The mean length of stay for elective admissions is 8.83 days (95% CI 8.66 to 9.01 days). Urgent admissions are associated with stays that are on average 26.8% (95% CI 21.7% to 32.1%) longer than those resulting from elective admissions. Emergency admissions result in stays that are on average 2.06 (95% CI 1.96 to 2.17) times as long as those from elective admissions.

However, note that all of these 95% CI rely on the Poisson assumption being correct and in fact this assumption is likely incorrect due to overdispersion as described above.

Question 5

Add age as factor variable to your model in question 4 and perform a likelihood ratio test of whether age (treated as categorical) is a predictor of length of hospital stay, after adjusting for type of admission. Use the reported deviance to infer whether this is an appropriate test.

SOLUTIONS

```
losPoissonTypeAge <- glm(los~type2+type3+factor(age),
                        data=medpar,family=poisson)
summary(losPoissonTypeAge)
```

Call:

```
glm(formula = los ~ type2 + type3 + factor(age), family = poisson,
    data = medpar)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.42738	0.11653	20.830	< 2e-16 ***
type2	0.23580	0.02097	11.247	< 2e-16 ***
type3	0.72507	0.02597	27.923	< 2e-16 ***
factor(age)2	-0.25416	0.12339	-2.060	0.03941 *
factor(age)3	-0.20133	0.11894	-1.693	0.09051 .
factor(age)4	-0.23440	0.11776	-1.990	0.04655 *
factor(age)5	-0.26663	0.11773	-2.265	0.02353 *
factor(age)6	-0.22920	0.11763	-1.949	0.05135 .
factor(age)7	-0.27528	0.11868	-2.320	0.02036 *
factor(age)8	-0.32308	0.12140	-2.661	0.00778 **
factor(age)9	-0.31245	0.12617	-2.476	0.01327 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 8901.1 on 1494 degrees of freedom
Residual deviance: 8165.2 on 1484 degrees of freedom
AIC: 13902

Number of Fisher Scoring iterations: 5

```
anova(losPoissonType,losPoissonTypeAge)
```

Analysis of Deviance Table

	Model 1:	Model 2:
	los ~ type2 + type3	los ~ type2 + type3 + factor(age)
	Resid. Df	Resid. Dev
1	1492	8183.6

```
2      1484      8165.2  8    18.442  0.01814 *
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The p-value from the likelihood ratio test is 0.018 - we have some evidence that there are differences in (log) mean length of stay between the different age groups after adjusting for admission type.

But the residual deviances in both models are much larger than the residual d.f. As such, the likelihood ratio test may not be reliable here for assessing whether there is an effect of age group conditional on admission type. We can however take the small p-value as evidence that the Poisson model with just admission type as covariate is not correctly specified.

Question 6

Use the `glm.nb` function in the `MASS` package to fit negative binomial models analogous to those you fitted in questions 4 and 5. Use `anova` to perform a likelihood ratio test. Compare your result with the previous question's findings.

SOLUTIONS

```
library(MASS)
losNBType <- glm.nb(los~type2+type3,data=medpar)
losNBTypeAge <- glm.nb(los~type2+type3+factor(age),
                      data=medpar)
anova(losNBType,losNBTypeAge)
```

Likelihood ratio tests of Negative Binomial Models

Response: los

	Model	theta	Resid. df	2 x log-lik.	Test	df
1	type2 + type3	2.233022	1492	-9600.438		
2	type2 + type3 + factor(age)	2.241159	1484	-9595.596	1 vs 2	8
	LR stat.	Pr(Chi)				
1						
2	4.841283	0.7743976				

We now have a p-value 0.77 for testing whether the age groups differ after conditioning on type of admission. This is a completely different result from the Poisson likelihood ratio test comparison from the previous question. The explanation is that the negative binomial model allows for the possibility of overdispersion.

Question 7

Write a short paragraph (suitable for a medical journal) to summarise your findings concerning the effects of the type of admission on length of stay in hospital based on the negative binomial model which includes type of admission as covariate but not age group.

SOLUTIONS

```
exp(coef(losNBType))
```

(Intercept)	type2	type3
8.830688	1.267877	2.065477

```
exp(confint.default(losNBType))
```

	2.5 %	97.5 %
(Intercept)	8.453975	9.224187
type2	1.149033	1.399012
type3	1.780669	2.395838

Urgent admissions were associated with stays that were on average 1.27 (95% CI 1.15 to 1.40) times longer than those resulting from elective admissions, while emergency admissions were associated with stays that were on average 2.07 (95% CI 1.78 to 2.40) times as long.

*Note that if we wanted an overall p -value testing the null that length of stay varied by type of admission, we could perform a likelihood ratio test of the model with **type2** and **type3** as covariates with the null model (only an intercept).*

Question 8

Use the estimates from your negative binomial model that includes only type of admission to calculate the estimated means and variances for each type of admission. Then compare these with the sample means and SDs for `los` in the three admission types. What do you conclude about the model?

SOLUTIONS

From the notes/slides, in the negative binomial model we have that if the expected (i.e. mean) count is μ , the variance is equal to $\mu(1 + \mu/\theta)$. We can thus calculate the required values with:

```
#estimated means are
mu0 <- exp(coef(losNBType)[1])
mu1 <- exp(coef(losNBType)[1]+coef(losNBType)[2])
mu2 <- exp(coef(losNBType)[1]+coef(losNBType)[3])

mu0
```

```
(Intercept)
  8.830688
```

```
mu1
```

```
(Intercept)
 11.19623
```

```
mu2
```

```
(Intercept)
 18.23958
```

```
#theta estimate is
losNBType$theta
```

```
[1] 2.233022
```

```
#estimated variances are
mu0*(1+mu0/losNBType$theta)
```

```
(Intercept)
 43.75244
```

```
mu1*(1+mu1/losNBType$theta)
```

```
(Intercept)
 67.33337
```

```
mu2*(1+mu2/losNBType$theta)
```

```
(Intercept)  
167.2226
```

You should ignore the '(Intercept)' label is printing in the above - this is just being carried through from the name of `coef(losNBType)[1]`.

We can then compared these with the sample means and variances in the three admission types. Here we'll do this using some functions from `dplyr`, but you can obviously also achieve the same with base R functions:

```
library(dplyr)  
medpar <- medpar %>%  
  mutate(  
    admission_type = factor(  
      case_when(  
        type1 == 1 ~ "Elective",  
        type2 == 1 ~ "Urgent",  
        type3 == 1 ~ "Emergency"  
      ),  
    levels = c("Elective", "Urgent", "Emergency")  
  )  
)  
  
medpar %>%  
  group_by(admission_type) %>%  
  summarise(  
    mean_los = mean(los),  
    var_los = var(los),  
    n = n()  
  )
```

```
# A tibble: 3 x 4  
  admission_type mean_los var_los    n  
  <fct>          <dbl>   <dbl> <int>  
1 Elective      8.83    41.7  1134  
2 Urgent       11.2    77.9   265  
3 Emergency    18.2   425.    96
```

The sample variances are close to those predicted by the negative binomial model for the elective and urgent admissions. However, for the emergency admissions the sample variance is quite a bit larger than predicted by the model. The number of emergency admissions is relatively small, meaning that the variance will be imprecisely estimated but the lack of agreement between the fitted and observed variances could also indicate that the Gamma random-effects distribution is not appropriate for these data (or that a different θ parameter is needed for each type of admission).

Another issue that we have overlooked in our analyses is that the hospital days only take positive values, whereas Poisson distributions (and negative binomials) should also sometimes take the value zero. One approach which could be used to allow for this is to instead model “number of days of stay after day of admission”, which would take the value zero for some patients.